

Main Manuscript for

**Antarctic Ozone Depletion Shapes Surface Cooling Pattern and
Climate Sensitivity**

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This PDF file includes:

Main Text
Figures 1 to 4

Abstract

Changes in sea surface temperature (SST) patterns are recognized as major feedbacks affecting climate sensitivity to increases in greenhouse gases. Over recent decades, while most of Earth's surface warmed, the eastern tropical Pacific and Southern Ocean unexpectedly cooled. These regional SST cooling trends are not quantitatively reproduced in global climate models (GCMs), contributing to biases in estimates of effective climate sensitivity. Although Antarctic ozone depletion has been proposed as a potential driver, its influence has generally been considered too weak to account for the observed cooling, in part because poorly constrained long-timescale ocean adjustment in GCMs can offset the initial cooling response. Here, we exploit internally generated interannual variability in Antarctic total-column ozone and global SST using a multiple linear regression model trained on ~4,000 years of unforced control simulations from eight GCMs. This approach isolates the robust short-timescale coupling between Antarctic ozone and SST while minimizing the influence of highly uncertain long-timescale ocean adjustment. Although this estimate represents a physically motivated upper bound on ozone's contribution, the inferred ozone-driven SST closely resembles observations in both spatial pattern and magnitude. Using a Green's function approach, we estimate that this ozone-driven SST pattern effect can make global radiative feedback more negative by 8-19% ($0.2\text{-}0.7\text{ W m}^{-2}\text{ K}^{-1}$) during 1979-2000, the period of most rapid ozone depletion, thereby reducing effective climate sensitivity. As Antarctic ozone recovery began in the 2000s, this pattern effect started to wane, reducing its stabilizing influence on global radiative feedback and increasing effective climate sensitivity.

Significance Statement

Antarctic ozone depletion can drive Southern Ocean sea surface temperature (SST) cooling, but its contribution over the past four decades has generally been considered too small to explain the observed SST cooling pattern. Coarse-resolution climate models typically simulate an initial cooling followed by a highly uncertain delayed warming from ocean adjustment. Some recent high-resolution climate models suggest this delayed warming may be weaker or absent, but their computational cost limits robust quantification of ozone's contribution. By isolating the robust short-timescale response, we show that ozone depletion can contribute substantially to the observed SST cooling, with the inferred response closely resembling observations in both spatial pattern and magnitude. This ozone-driven SST pattern effect helps explain historical changes in effective climate sensitivity.

Main Text

Introduction

Although global climate models (GCMs) capture the historical rise in global-mean surface temperature, they fail to reproduce some key spatial features—most notably the cooling trends in the eastern tropical Pacific and the Southern Ocean that have persisted over the past four decades (1) (Fig. 1). This discrepancy between GCMs and observations exemplifies the SST “pattern effect”,

in which the spatial pattern of SST influences the strength of climate feedbacks and, in turn, the rate of global warming (2, 3). In particular, cooling in the eastern tropical Pacific promotes low-cloud cover and enhances shortwave reflection, reducing net radiative heating of the climate system (4). Given the observed SST trend pattern, recent decades exhibit more stabilizing climate feedbacks and are thus less prone to warming, whereas GCMs lack this cooling pattern on average and therefore tend to have less stabilizing climate feedbacks (5). In addition, the observed eastern tropical Pacific cooling may have influenced climate in remote regions through atmospheric teleconnections, such as drought in the western United States (6). Understanding the observed SST cooling pattern is thus key for interpreting historical climate change, assessing global climate sensitivity, and improving future global and regional climate projections. However, the mechanisms responsible for this cooling signature remain uncertain (7).

Antarctic ozone depletion has been proposed as a potential driver of Southern Ocean SST cooling by strengthening and shifting the westerlies at high latitudes and thereby altering wind-driven ocean circulation (8). This poleward jet shift characterizes a transition to the positive phase of the Southern Annular Mode (SAM) since the 1980s, driven largely by Antarctic ozone depletion (9, 10). Despite this well-established mechanistic link, evidence from empirical studies based on linear regression analysis (11), GCM experiments with prescribed ozone forcing (12), and studies that convolve the Southern Ocean response with observed ozone changes using idealized GCM ozone perturbation experiments (13) suggests that ozone depletion explains less than 10% of the observed Southern Ocean SST cooling in recent decades.

However, these approaches have limitations that may lead to an underestimate of ozone's contribution. Decadal-scale variability of Southern Ocean SSTs is also influenced by the response of the SAM to greenhouse gases (9, 14, 15) and is further complicated by remote influences from tropical Pacific SSTs via atmospheric teleconnections (16, 17). As a result, regressions based on short observational records may confound ozone's effects with other factors. In addition, standard-resolution GCMs (e.g., 1° horizontal resolution) tend to simulate a spurious enhancement of poleward ocean heat transport in response to an increase in the SAM on decadal timescales, inconsistent with both observations and results from high-resolution (e.g., 0.1° horizontal resolution) eddy-resolving ocean models (18–21). This long-timescale warm bias arising from the lack of resolved ocean eddies in the Southern Ocean could substantially dampen the ozone-depletion-induced cooling signal in standard-resolution GCMs, leading to an underestimate of ozone's contribution. In principle, eddy-resolving ocean models could better represent the long-timescale Southern Ocean SST response to Antarctic ozone depletion. However, ozone-only single-forcing initial-condition large ensembles (22) with a sufficient number of members to cleanly isolate the forced response from internal variability have not yet been conducted with coupled ocean eddy-resolving models (23) due to their prohibitive computational cost. These limitations motivate isolating the robust short-timescale response from the spurious long-timescale response

in standard-resolution GCMs to constrain an upper limit of the Southern Ocean SST response to Antarctic ozone changes.

SST response to Antarctic ozone depletion

To characterize the simulated short-timescale intrinsic relationship between Antarctic ozone and SST, we construct a multiple linear regression (MLR) model in which annual-mean SST at each grid point is regressed onto the preceding year's (lag 1-year) September-December polar-cap (60°S-90°S) total column ozone using ~4,000 years of pre-industrial control (PiControl) simulations from eight GCMs archived in Phase 6 of the Coupled Model Intercomparison Project (CMIP6; Table S1 and Methods). The selected GCMs employ either simplified or fully interactive stratospheric chemistry (24). Therefore, despite their fixed pre-industrial levels of ozone-depleting substances, stratospheric ozone exhibits internally generated variability consistent with two-way coupling between ozone and atmospheric dynamics.

The advantage of our methodology compared to regressions based on either observations or historical simulations is that the long PiControl simulations eliminate time-varying external forcings and span many phases of natural variability, enabling a rich sampling of the simulated ozone-SST relationship. This approach also isolates the surface temperature response to Antarctic ozone changes from the direct radiative forcing effects of global ozone and the ozone-depleting substances themselves. The disadvantage is that it relies on the fidelity of the simulated relationships between Antarctic ozone and SST, as well as on the assumption that the relationships derived from PiControl simulations remain valid under present-day conditions. Acknowledging GCM-specific biases, we establish robustness by considering only regions where all eight GCMs agree on the sign of the SST response to Antarctic ozone, with the spread of response magnitudes in these areas providing an estimate of inter-model uncertainty. Regression coefficients from individual GCMs and the multi-model-means for each predictor in the MLR are shown in Figs. S1-3. Details of the MLR and additional sensitivity tests are provided in the Methods section and Fig. S4, including training the MLR on historical simulations or observations to assess whether the PiControl ozone-SST relationship holds under present-day conditions, and tests of ozone seasonality and its impact on SST.

To quantify SST responses to historical changes in Antarctic ozone, we apply the multi-model-mean intrinsic ozone-SST relationship derived from PiControl simulations to the satellite-observed Solar Backscatter Ultraviolet (SBUV) ozone dataset (25). We then compare the inferred SST response against three widely used observation-based SST reconstructions from COBE-SST2 (26), ERSSTv6 (27, 28), and HadISST1 (29). The pattern and magnitude of the ozone-driven SST cooling over 1979-2024 (Fig. 1C) closely resemble those of the mean of the three observational SST reconstructions (Fig. 1A), particularly in the eastern tropical Pacific and Southern Ocean. The difference between Figs. 1A and 1C, representing observed SSTs with the ozone-driven component removed, is shown in Fig. 1D. Averaged over the dashed contours in Figs. 1A-D, where all eight GCMs exhibit a robust cooling response to declining Antarctic ozone, the three

observational SST datasets yield cooling trends of -0.02 ± 0.04 K dec⁻¹ (COBE-SST2), -0.07 ± 0.04 K dec⁻¹ (ERSSTv6), and -0.04 ± 0.03 K dec⁻¹ (HadISST1) (95% confidence interval) during 1979–2024, significantly contrasting with the global-mean warming rate (Fig. 1E). The MLR model suggests that Antarctic ozone changes contribute substantially to the SST cooling (-0.07 ± 0.03 K dec⁻¹) over the same region and period. After removing this ozone-driven component, the residual SST trends in the eastern tropical Pacific and Southern Ocean are markedly more consistent with the global-mean warming. Note that the remaining discrepancy suggests that additional processes such as internal variability (30) also contribute substantially to the observed SST cooling.

Previous studies have suggested that Antarctic ozone depletion can produce a broad spatial pattern of cooling that resembles observations, but the simulated response is typically an order of magnitude weaker than observed (11, 12). This raises the question: why do the results of the MLR model, trained on the linkages between Antarctic ozone and SSTs in PiControl simulations, agree with the observed magnitude of eastern tropical Pacific and Southern Ocean SST cooling?

Studies using standard-resolution GCMs generally find that the Southern Ocean cools in response to an increase in the SAM on short (i.e., annual) timescales but warms on long (i.e., multi-annual and decadal) timescales. However, both the magnitude and timing of this delayed warming vary substantially across models (13, 15, 31). This behavior is evident in the climate response functions derived from companion simulations with both standard- and high-resolution models (Fig. S5; see Methods). Despite considerable inter-model spread, the high-resolution configurations of CESM1-CAM5, EC-Earth3P, and HadGEM3-GC31 exhibit a weaker (or no) delayed warming response relative to their standard-resolution counterparts. This reduction is likely associated with mesoscale eddy compensation, which suppresses the interior ocean upwelling of warm deep water (19). Nevertheless, delayed warming persists in some high-resolution simulations (32), indicating that the strength of mesoscale eddy compensation is model dependent, as also illustrated by ECMWF-IFS in Fig. S5. In contrast, the short-timescale cooling response is considerably more robust across both standard- and high-resolution simulations (33) and is supported by observations (19). By regressing SST onto Antarctic ozone from the preceding year, our MLR model primarily isolates this short-timescale ozone-SST coupling, thereby limiting the influence of long-timescale ocean adjustments that remain both uncertain and model dependent.

The short-timescale coupling between Antarctic ozone and SSTs in the PiControl simulations is evident in lead-lag composites of sea-level pressure (SLP) and 500-hPa geopotential heights (Z500) between low (<5th percentile) and moderate (40th–60th percentile) Antarctic ozone years (Fig. 2). Antarctic ozone variability in PiControl simulations arises primarily from the El Niño–Southern Oscillation (ENSO): El Niño/La Niña accelerates/slows the Brewer-Dobson circulation in the stratosphere, transporting more/less ozone-rich air poleward (34), with ENSO leading Antarctic ozone by ~12 months (35). Consistent with this mechanism, composite SST anomalies one year before low Antarctic ozone show pronounced La Niña-like cooling in the tropical Pacific (Fig. 2A). The accompanying SLP and Z500 anomaly patterns—enhanced west-east tropical Indo-Pacific SLP gradient and broad tropical-mean reduction in Z500 (Figs. 2D,G)—likewise reflect the canonical ENSO-troposphere coupling (36, 37). Once La Niña establishes anomalously low

Antarctic ozone during austral spring (September-December), ozone exerts its strongest influence on Southern Hemisphere climate in the following summer (January-April). The resulting patterns resemble the positive phase of the SAM, featuring stronger meridional SLP and Z500 gradients associated with a poleward-shifted jet that intensifies equatorward Ekman heat transport, leading to SST cooling in the Southern Ocean (9) (Figs. 2B,E,H). Cold air originating over the Southern Ocean is advected equatorward and subsequently amplified and sustained by positive feedbacks in both the atmosphere and the ocean along the west coast of South America, thereby reinforcing cooling in the southeast and eastern tropical Pacific (38, 39).

Impact on historical climate sensitivity

“Pattern effects” associated with the spatial structure of SST trends, including cooling in the eastern tropical Pacific and Southern Oceans have slowed global warming during the historical period (5, 40). To quantify the contribution of Antarctic ozone depletion and recovery to these pattern effects, we apply a Green’s function approach (4, 41, 42) to estimate the temporal evolution of the global radiative feedback parameter associated with the ozone-induced component of observed SST trends (see Methods). Fig. 3 shows the percent contribution from ozone-induced changes in the pattern of SST trends for top-of-atmosphere net radiation (R ; negative values indicate outgoing radiation), global-mean surface temperature (T), and the global radiative feedback parameter ($\lambda = R/T$) computed from sliding 22-year SST trend windows. The corresponding absolute contributions are shown in Fig. S6.

During 1979-2000, when Antarctic ozone depletion was strongest, the resulting ozone-induced SST cooling based on the MLR model was also largest in the eastern tropical Pacific (Fig. S7B), where radiative feedbacks are particularly sensitive (4, 41–43). We estimate that the ozone-induced SST trend pattern during this period increased radiative cooling to space by 2-6% (6.4-20.4 mW m⁻² dec⁻¹), with the range representing the spread in mean estimates across different Green’s functions (see Methods). At the same time, the ozone-induced SST trend pattern reduced the rate of global-mean surface warming by 7-18% (0.01-0.02 K dec⁻¹). Together, these effects produced an 8-19% (0.21-0.70 W m⁻² K⁻¹) strengthening of the global radiative feedback (i.e., making λ more negative). In other words, the SST trend pattern induced by ozone depletion enhanced radiative damping, leading to more energy lost to space per degree of surface warming and a more stable, less warming-prone climate state.

When the analysis is extended to 1979-2014 and 1979-2022, periods commonly examined in previous studies (7, 12, 40), during which the effects of ozone depletion continue to outweigh those of ozone recovery and thus produce an SST cooling pattern, the ozone-driven SST pattern effect

contributes to a strengthening in global radiative feedback by 3-7% ($0.09\text{-}0.24 \text{ W m}^{-2} \text{ K}^{-1}$) and 2-5% ($0.06\text{-}0.15 \text{ W m}^{-2} \text{ K}^{-1}$), respectively.

SST pattern effect under ozone recovery

The ozone-induced SST cooling in the eastern tropical Pacific weakens during the 21st-century ozone recovery period (Fig. S7B). As a result, ozone's contribution to enhanced radiative cooling correspondingly diminishes, and global-mean surface warming accelerates. This shift yields a less stable, more warming-prone climate state, reflected in a positive change in λ for periods beginning after the mid-1990s (Fig. 3). Notably, the time-varying climate sensitivity we diagnose from ozone-induced SST pattern changes closely matches the behavior reported in (5), which identifies a substantially lower-sensitivity state emerging after the 1980s relative to that expected from long-term CO₂ increases, followed by a gradual waning of this difference after the 2000s.

Due to declines in ozone-depleting substances, signs of Antarctic ozone recovery have already emerged in the 21st century (44, 45). However, because of the large natural variability in ozone and the short observational record, along with the exceptional 2020 Australian wildfires and the 2022 Hunga-Tonga volcanic eruption that contributed to additional ozone losses (46, 47), the SSTs explained by observed ozone according to our MLR method still show a weak cooling trend in the eastern tropical Pacific and Southern Ocean during 2000-2024 (Fig. 4A). In contrast, GCM-simulated ozone, which combines historical and scenario (SSP2-4.5) forcings and does not include the recent exceptional wildfire- and volcano-driven ozone losses (see Methods), yields a reversal in sign of the MLR-derived SST trends during 2000-2024 (i.e., a warming pattern in the eastern tropical Pacific and Southern Ocean; Figs. 4B-C).

Under the SSP2-4.5 scenario, Antarctic ozone is projected to return to 1980 levels by the middle of the 21st century (24), implying that the SST cooling in the eastern tropical Pacific and Southern Ocean induced by ozone depletion and the associated shift toward a more negative global radiative feedback should likewise subside by the 2050s, all other factors being equal (Fig. 4C). Note that the SST projections in Fig. 4C reflect only the response to changes in Antarctic ozone. As ozone recovery proceeds more gradually than the preceding rapid depletion, future changes in other factors such as anthropogenic aerosols (48), greenhouse gases (49), and Antarctic ice-sheet melt (50) may increasingly dominate SST evolution and influence the timing of the "de-emergence" of the SST pattern effect associated with Antarctic ozone recovery.

Discussion

We provide numerical evidence suggesting that Antarctic ozone depletion can contribute substantially to the observed SST cooling in the eastern tropical Pacific and Southern Ocean. This evidence is based on a MLR model developed from the statistics of interannual ozone-SST dynamical coupling in unforced (PiControl) simulations, which avoids biases arising from

unresolved ocean eddy adjustments on longer timescales in standard-resolution GCMs. We find that both the spatial structure and magnitude of this short-timescale intrinsic ozone-SST relationship are robust across PiControl simulations, historical simulations, and observations. However, because long-timescale responses remain highly model dependent even in eddy-resolving ocean models (19–21, 32), and observational records are too short to constrain their magnitude (19), our focus on the short-timescale response should be interpreted as a physically motivated upper-bound estimate of ozone-driven SST changes.

We also estimate that this ozone-depletion-induced SST trend pattern has contributed to a decrease in global climate sensitivity in recent decades, with the strongest effect during 1979-2000, strengthening the global radiative feedback by 8-19%. Likewise, if Antarctic ozone continues to recover from its minimum around the turn of the 21st century, the global radiative feedback is expected to shift toward a higher-sensitivity, more warming-prone climate. In this context, it is important to emphasize that the climate sensitivity results presented here isolate only the ozone-induced SST pattern effect and do not represent the full range of radiative, chemical, and dynamical benefits associated with ozone recovery, which remain essential for climate and environmental protection. More broadly, these findings highlight Antarctic ozone as a key regulator of global climate through its impact on SST pattern effects, and underscore the importance of representing chemistry-climate interactions in the coupled stratosphere-troposphere-ocean system for improved understanding of historical and future climate change.

Materials and Methods

Observation and model descriptions

We consider three widely used, global, gridded SST datasets from 1979 to 2024: the Japan Meteorological Agency Centennial Observation-Based Estimates of SSTs (COBE-SST2) (26), the U.S. National Oceanic and Atmospheric Administration Extended Reconstructed SST (ERSSTv6) (27, 28), and the U.K. Met Office Hadley Centre Sea Ice and Sea Surface Temperature dataset (HadISST1) (29). These datasets are reconstructed primarily from *in situ* ship and buoy measurements, with HadISST1 further supplemented by satellite observations after the 1980s. The Southern Ocean remains the region with the sparsest observational coverage, even after 1980 (51), resulting in substantial inter-dataset uncertainties. To account for this observational uncertainty, we include all three reconstructed SST products in our analysis.

We consider satellite observations of total column ozone from the Solar Backscatter Ultraviolet (SBUV) version 8.7 (25), which provide a monthly and zonally averaged ozone dataset from 1970 to 2023. We use the period from 1978 to 2023 to estimate ozone-explained SSTs for the period from 1979 to 2024. The SBUV observations have been validated against ground-based Dobson and Brewer measurements and are widely used in past ozone assessments to characterize long-term global and polar ozone changes (52, 53).

The models used to construct the MLR in this study are based on PiControl simulations from eight CMIP6 GCMs that employ either simplified or fully interactive stratospheric chemistry schemes. For consistency, we use 499 years from each GCM, although some (e.g., MRI-ESM2-0 and UKESM1-0-LL) provide longer PiControl simulations. To remove any potential nonlinear model drifts in the PiControl simulations, all variables are detrended by fitting and removing a second-order polynomial from each monthly time series at each grid point. We also use one realization from each GCM's historical simulation (up to 2014) and the SSP2-4.5 scenario (after 2014) in certain analyses (Figs. 1E and 4B-C). Since two of the eight GCMs do not provide SSP simulations, the multi-model-mean after 2014 is calculated from the remaining six models. For the MLR sensitivity test shown in Fig. S4F, we use all realizations from the historical simulations of CNRM-CM6-1, CNRM-ESM2-1, MRI-ESM2-0, and UKESM1-0-LL for the period 1975-2014. The list of GCMs used is summarized in Table S1, and additional details, including their respective chemistry schemes, are provided in (24).

Unless otherwise specified, all spatial averages presented in this study are weighted by the cosine of latitude to account for variations in grid-cell area across latitude.

Multiple linear regression and sensitivity tests

We construct the following MLR for each CMIP6 GCM using its 499 years of PiControl simulations to characterize the intrinsic relationship between Antarctic ozone and SST:

$$SST(i, j, t) = \alpha(i, j) \cdot Ozone(t - 1) + \beta_1(i, j) \cdot ENSO(t - 1) + \beta_2(i, j) \cdot ENSO(t) + \epsilon \quad (1)$$

where the annual-mean SST anomaly at each grid point (i, j) in year t is regressed onto September-December total column ozone anomalies averaged over 60°S-90°S from the preceding year ($t-1$), and the ENSO indices, calculated as annual-mean SST anomalies averaged over the Niño 3.4 region (5°N-5°S, 170°W-120°W) in years $t-1$ and t . Integrated total column ozone is used here because it primarily reflects ozone variability in the lower stratosphere, consistent with the altitude range where present-day ozone depletion dominates (54). Accordingly, the surface impact associated with lower-than-normal column ozone in PiControl simulations exhibits behavior consistent with our understanding of the surface response to present-day Antarctic ozone depletion (9) (Fig. 2). The observed Antarctic ozone change over the past four decades falls reasonably within the distribution of internal variability simulated in the PiControl runs (Fig. S8). ENSO indices are included as additional predictors because Antarctic ozone variability is partly modulated by ENSO teleconnections (55), which also influence global SST directly, with ENSO leading SST by about 0-1.5 years (56). Including both lag 1-year and lag 0-year ENSO indices allows the partial regression coefficient on Antarctic ozone to capture, as far as possible, the local ozone influence on SST.

Given differences in model physics, some GCMs (e.g., CNRM-CM6-1 and E3SM-1-0) exhibit a stronger SST response to Antarctic ozone, whereas others (e.g., GFDL-ESM4) show a weaker

response, as illustrated by the partial regression coefficients for Antarctic ozone in individual GCMs (Figs. S1A-H). Our analysis focuses on regions where all eight GCMs agree on the sign of the SST response to Antarctic ozone, highlighted by dashed contours in the multi-model-mean (Fig. S1I). Applying this criterion, the eastern tropical Pacific and the Southern Ocean both emerge as regions with a robust SST response to Antarctic ozone variability. In both regions, all GCMs yield positive coefficients, indicating SST cooling with Antarctic ozone depletion (i.e., lower ozone). The warming in the South Indian Ocean (in response to a decrease in Antarctic ozone) and the dipole pattern in the South Atlantic Ocean are also consistent with SAM-based regression from observations (8). Importantly, this robust inter-model agreement between Antarctic ozone and SST is largely confined to the Southern Hemisphere, suggesting that ENSO teleconnections are effectively minimized in the partial regression on Antarctic ozone.

We conduct additional sensitivity tests to examine the robustness and physical linkage between Antarctic ozone and SST. The key relationship between Antarctic ozone depletion during springtime and lower-than-normal SSTs in the eastern tropical Pacific and the Southern Ocean is both robust across different analysis designs and consistent with physical expectations. For instance, similar patterns and magnitudes emerge in MLR analyses using Antarctic ozone averaged in September-October (Fig. S4A) and November-December (Fig. S4B). In contrast, this relationship disappears when using Antarctic ozone averaged in March-April (Fig. S4C), when the stratospheric polar vortex has largely broken down and ozone variability is no longer dynamically coupled to the tropospheric circulation (57). We also test a simple linear regression between Antarctic ozone and SST (without explicitly including the ENSO terms), and the magnitude of the cooling over the eastern tropical Pacific and the Southern Ocean remains comparable to that obtained from the MLR (Fig. S4D), underscoring the robustness of the ozone-SST relationship in these regions.

We further test whether the PiControl ozone-SST relationship holds under present-day conditions by constructing an MLR using both observations (based on data from 1979-2022) and CMIP6 historical simulations (based on model outputs from 1975-2014). Because observations can conflate forced responses and internal variability, we first remove the forced component from observed SST, estimated as the average of the top five most skillful methods evaluated in the Forced Component Estimation Statistical Method Intercomparison Project (ForceSMIP) (30). We then apply the MLR between observed ozone (from SBUV) and the SST internal variability (with the forced component removed from ERSSTv5), after first removing second-order polynomial trends from both variables to largely isolate their interannual variability. The resulting ozone-driven SST trend is shown in Fig. S4E. Additionally, we construct the MLR based on internal variability in a subset of four GCMs from Table S1 (CNRM-CM6-1, CNRM-ESM2-1, MRI-ESM2-0 and UKESM1-0-LL), each of which provides at least 10 historical realizations. By removing the ensemble mean from each realization, the residual largely reflects internal variability. The resulting SST trend from MLR based on these historical simulations is shown in Fig. S4F. Hatching in the eastern tropical Pacific and Southern Ocean in Figs. S4E-F indicates that the ozone-driven cooling derived from internal variability in either observations or historical simulations is consistent with the PiControl ozone-SST relationship, falling within the $\pm 1\sigma$ inter-model uncertainty.

Climate response function calculation

Following (13, 33), we estimate the Southern Ocean SST response to a step increase in the SAM index using control simulations from five GCMs participating in the High Resolution Model Intercomparison Project (HighResMIP) (58). These GCMs provide paired standard- and high-resolution configurations, each spanning approximately 100 years, enabling a comparison of SST responses across resolutions without confounding differences in model physics. All model outputs are detrended by removing a second-order polynomial fit from each monthly time series to account for potential nonlinear model drift.

The Southern Ocean SST anomalies can be represented as a convolution of impulse response functions G with respect to the SAM index. A discretized form of this integral is given by:

$$SST(t) \approx \sum_{\tau=0}^{\tau_{max}} G(\tau) \cdot SAM(t - \tau) \cdot \Delta\tau + \epsilon \quad (2)$$

where the Southern Ocean SST is the annual-mean SST averaged over 50°S-70°S, and SAM is the December-May averaged SAM index, calculated as the monthly normalized SLP difference between 40°S and 65°S. These definitions follow (13), which shows that SST responses to SAM estimated in this way provide a good proxy for the responses to Antarctic ozone depletion. The lag τ is measured in years with a time increment $\Delta\tau = 1$ year. The response function G is then estimated using MLR of Southern Ocean SST anomaly onto SAM indices at various time lags. To reduce sensitivity to fitting choices, we average results across a range of fitting parameter spaces (33). Specifically, the maximum lag cutoff τ_{max} is varied from 50 to 80 years in 1-year increments, and the full time series is truncated by 0-8% at both the beginning and end in 2% increments. The climate response function (CRF) of the Southern Ocean SST t years after a 1σ step increase in SAM is then estimated as:

$$CRF(t) = \sum_{\tau=0}^t G(\tau) \cdot \Delta\tau \quad (3)$$

Earlier studies calculating CRFs from standard-resolution GCMs generally find that Southern Ocean SST cools in response to a positive SAM anomaly on short timescales but warms on longer timescales, although the magnitude of long-term warming is highly uncertain and model dependent (13, 15, 33). Fig. S5 shows CRFs in both the high- and standard-resolution configurations for each HighResMIP GCM, illustrating how the Southern Ocean SST response changes with increased model resolution. For CESM1-CAM5 and HadGEM3-GC31, whose high-resolution ocean grids are finer than 0.1°, the minimum resolution required to fully resolve mesoscale ocean eddies (59), their high-resolution configurations exhibit substantially weaker (or no) long-timescale warming than their standard-resolution counterparts. Although CNRM-CM6-1 and ECMWF-IFS show mixed or

even opposite behavior, their high-resolution configurations adopt a 0.25° ocean grid, which may still be too coarse to fully resolve mesoscale ocean eddies (59, 60).

Green's function approach for climate sensitivity analysis

The Green's function approach enables us to quantify how a given SST pattern modulates the global climate feedback (4, 41, 42), defined as the global radiative response per degree of global-mean surface temperature change:

$$\lambda = \frac{R}{T} = \frac{\sum_{i,j} GF_R(i,j) \cdot SST_{trend}(i,j)}{\sum_{i,j} GF_T(i,j) \cdot SST_{trend}(i,j)} \quad (4)$$

where R and T represent the trends in top-of-atmosphere net radiation (units in $\text{W m}^{-2} \text{dec}^{-1}$) and global-mean surface temperature (units in K dec^{-1}), obtained by integrating the Green's functions for radiation (GF_R) and surface temperature (GF_T) with a given SST trend pattern (SST_{trend}) globally. The resulting λ (units in $\text{W m}^{-2} \text{K}^{-1}$) represents the global radiative feedback associated with the SST trend pattern over the given trend period.

The Green's functions used in this study are derived from CESM1-CAM4 (41), CESM1-CAM5 (4), and GFDL-AM4 (42) to estimate the range of uncertainty arising from differences in model physics. To ensure consistent spatial integration, all datasets (including observations, CMIP6 outputs, and the Green's functions) are first re-gridded to the horizontal resolution of CESM1-CAM4 (approximately $1.9^\circ \times 2.5^\circ$). Note that sea ice pattern effects can also play an important role in the global energy budget (61). Among the models considered here, only CESM1-CAM4 includes the effects of sea ice changes (41), whereas the others mainly prescribe sea ice. This difference may partly explain why the changes in R , T , and λ associated with Antarctic ozone are more pronounced when using the Green's function from CESM1-CAM4 (Figs. 3 and S6).

We apply a sliding 22-year trend window from 1979 to 2024 to capture the gradual transition from the ozone-depletion era (mainly 1979-2000) to the ozone-recovery era (after 2000). We calculate R , T , and λ for the observed SST trend patterns using three observation-based SST datasets (COBE-SST2, ERSSTv6, HadISST1), representing real-world trends that include both external forcings and internal variability. For each observed SST dataset, we also subtract the ozone-explained SST component to estimate R , T , and λ representing a hypothetical world without ozone-induced SST changes. The subtraction is applied only over regions where all eight GCMs agree on the sign of the SST response to Antarctic ozone (dashed regions in Fig. S11), including robust SST cooling in the eastern tropical Pacific and the Southern Ocean and warming in the South Indian Ocean and South Atlantic Ocean due to Antarctic ozone depletion. The resulting differences in R , T , and λ between those derived from observed SST and those derived from observed SST with the ozone-explained component removed represent the contribution of Antarctic ozone changes. The relative contributions are shown in Fig. 3, and the absolute contributions are shown in Fig. S6.

Data, Materials, and Software Availability

CMIP6 PiControl, historical, SSP2-4.5, and HighResMIP model outputs are archived at the Earth System Grid Federation (<https://aims2.llnl.gov/>). COBE-SST2 is available from the NOAA Physical Sciences Laboratory (<https://psl.noaa.gov/data/gridded/data.cobe2.html>), ERSSTv6 is from the NOAA National Centers for Environmental Information (<https://www.ncei.noaa.gov/products/extended-reconstructed-ssst>), and HadISST1 is from the U.K. Met Office (<https://www.metoffice.gov.uk/hadobs/hadisst/>). ForceSMIP SST data are available on Zenodo (<https://doi.org/10.5281/zenodo.17106679>). The SBUV ozone dataset is available from the NASA Goddard Space Flight Center (https://acd-ext.gsfc.nasa.gov/Data_services/merged/). All CMIP6 and observational SST datasets are re-gridded to a common horizontal resolution, and the re-gridded datasets and code used in this study are available at Zenodo (<https://doi.org/10.5281/zenodo.22267256>).

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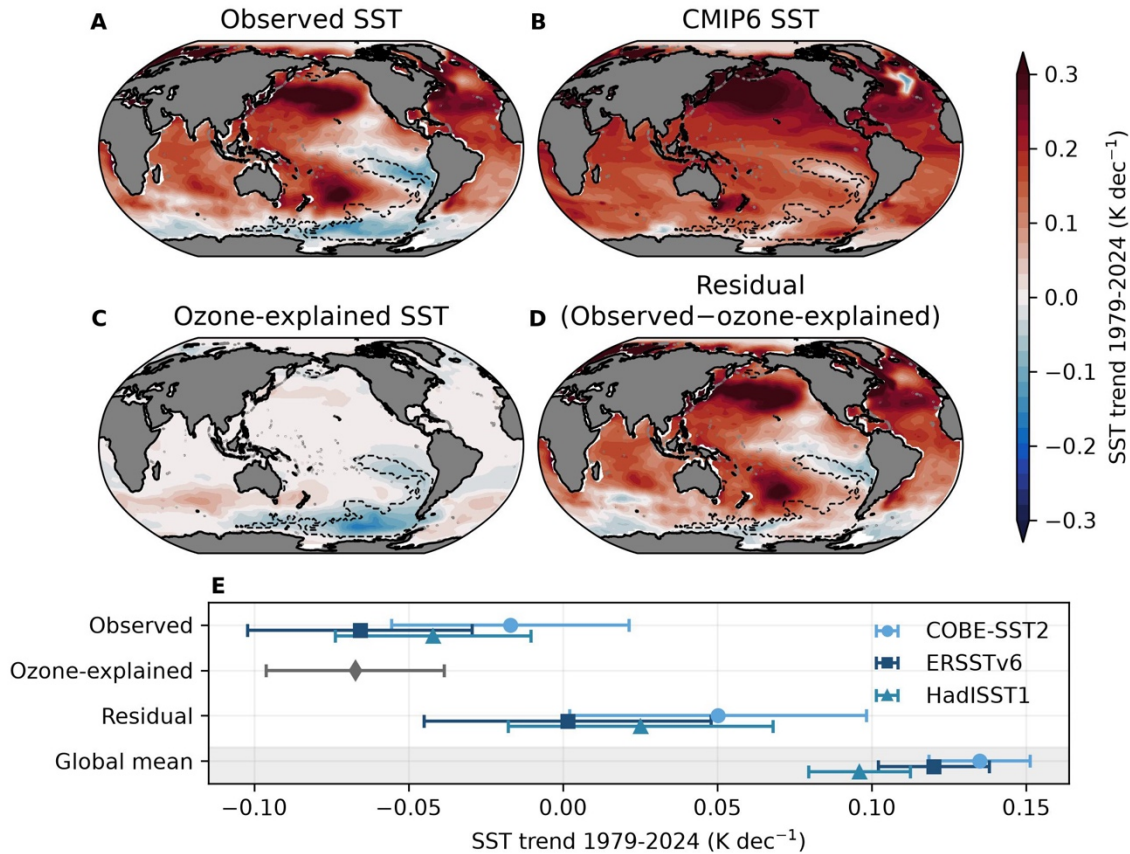


Fig. 1. Comparison of observed and modeled sea surface temperature trends. Panels A-D show linear SST trend maps from 1979 to 2024. Panel A shows observations averaged from three SST datasets (COBE-SST2, ERSSTv6, and HadISST1). Panel B shows the multi-model-mean direct SST output from eight CMIP6 GCMs (listed in Table S1). Panel C shows the multi-model-mean ozone-explained SST from the MLR based on the same eight GCMs. Panel D shows the residual SST trend obtained by removing the ozone-driven component in C from the observed SST trend in A. Dashed contours in A-D denote regions where all eight GCMs in MLR exhibit SST cooling in response to Antarctic ozone depletion. Panel E shows SST trends averaged over the dashed-contour region for the observational datasets, ozone-driven SST response, and residual SST trend, together with the observed global-mean SST trend (in light gray shading). Error bars in E indicate 95% confidence intervals (± 2 standard errors).

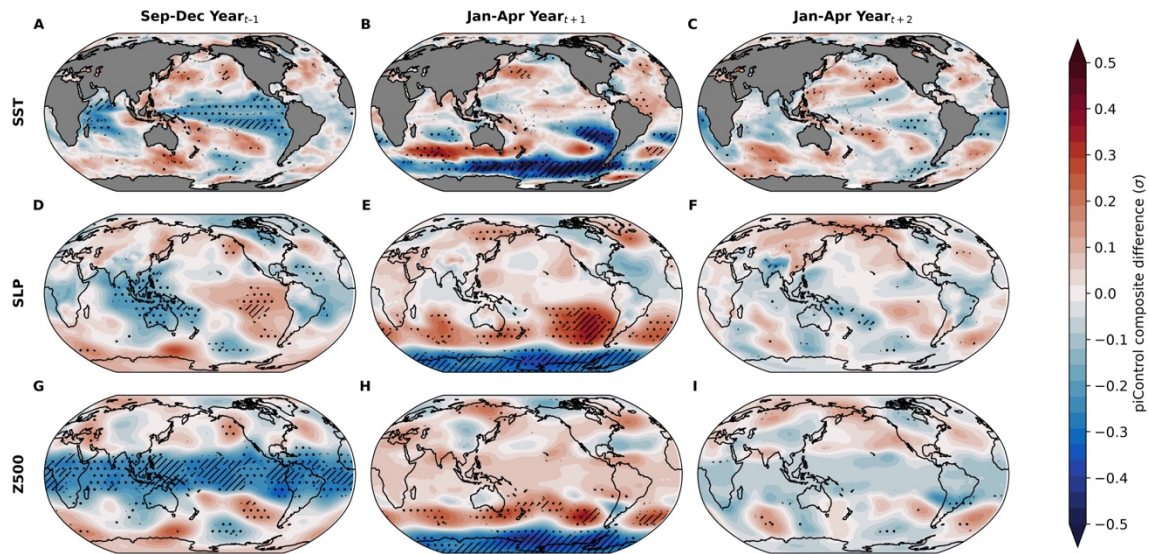


Fig. 2. Composite sea surface temperature and atmospheric circulation anomalies associated with low Antarctic ozone. Panels A-I show multi-model-mean composite differences between low Antarctic ozone years (<5th percentile) and moderate Antarctic ozone years (40th-60th percentile) in the eight CMIP6 PiControl simulations. Differences in SST (A,B,C), SLP (D,E,F), and Z500 (G,H,I) are shown for the austral spring one year prior to low Antarctic ozone (A,D,G), the subsequent austral summer (B,E,H), and the austral summer two years later (C,F,I). Each field is normalized by its own standard deviation within each GCM's PiControl run before averaging across the eight GCMs, resulting in anomalies in units of σ . Hatching indicates regions where all eight GCMs agree on sign, and dots mark regions where seven of eight GCMs agree on sign.

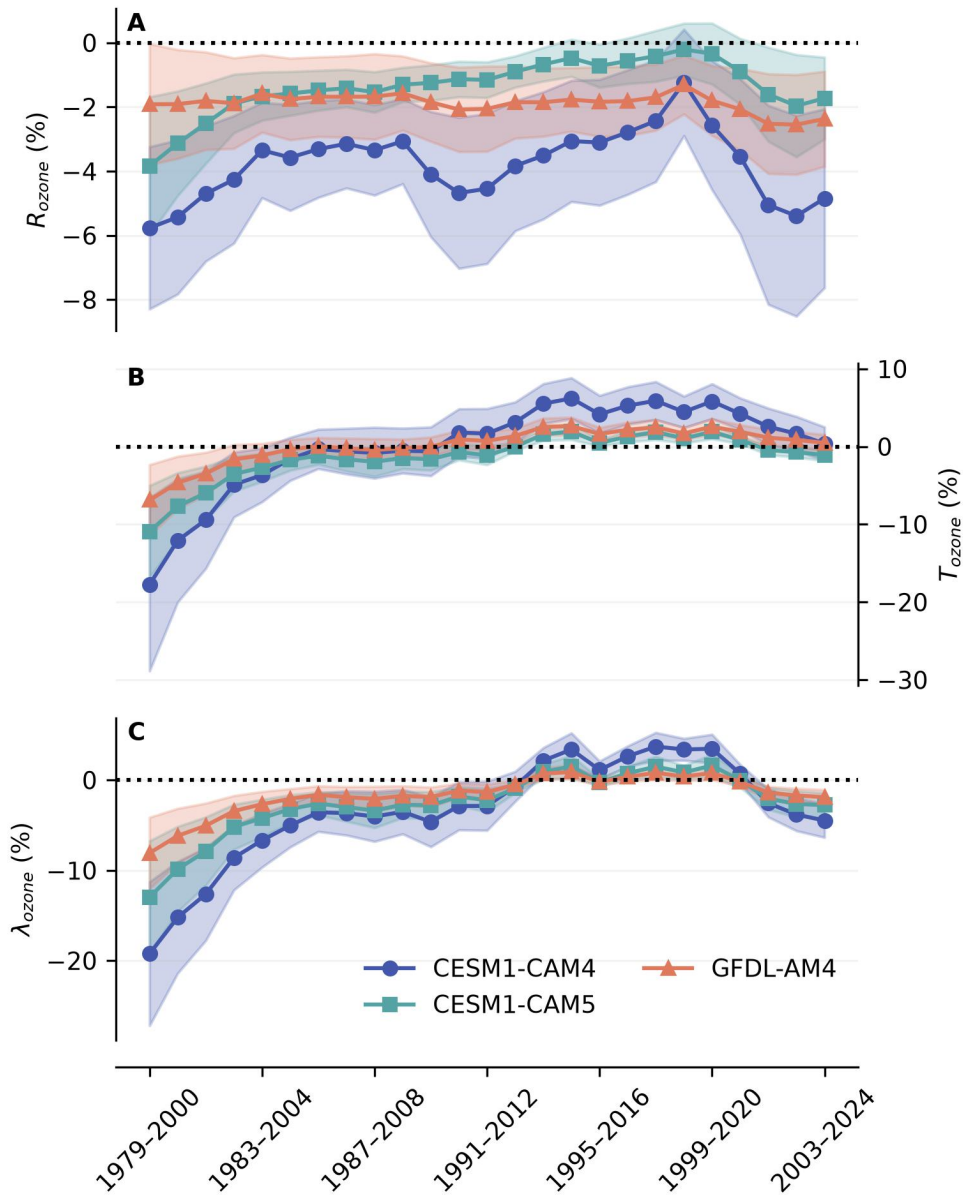


Fig. 3. Climate sensitivity due to ozone-induced sea surface temperature pattern changes.

Panel A shows percent difference in top-of-atmosphere net radiation R , panel B in global-mean surface temperature T , and panel C in global radiative feedback parameter λ , each estimated using the Green's function between observed SST and its counterpart with the ozone-explained SST removed. Accordingly, these percent differences quantify the contribution of ozone-induced SST pattern changes in R , T , and λ . Results are calculated over sliding 22-year trend windows from 1979 to 2024. Each color represents a different Green's function; the solid curve shows the mean estimate, and the shading denotes the $\pm 1\sigma$ uncertainty arising from the spread across the three observational SST products and the uncertainty in the MLR regression coefficients.

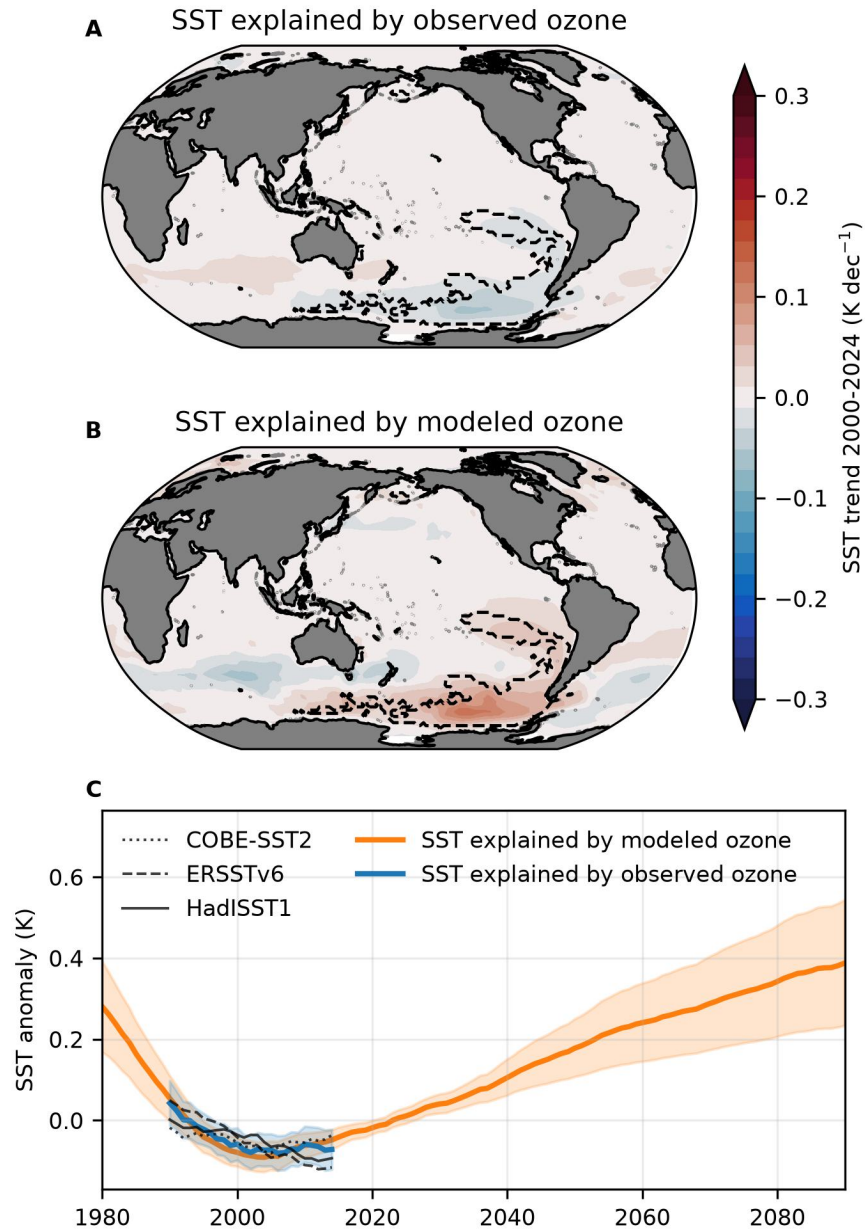


Fig. 4. Comparison of ozone-explained sea surface temperature using observed and model-simulated ozone. Panels A and B show SST trend maps from 2000 to 2024 derived from the MLR using observed ozone from SBUV and simulated ozone from CMIP6 (with historical simulations prior to 2014 and SSP2-4.5 thereafter), respectively. Panel C shows 22-year running-mean SST anomalies averaged over the eastern tropical Pacific and Southern Ocean (region enclosed by the dashed contours in A-B). Blue and orange curves in panel C denote SST driven by SBUV and CMIP6 ozone, respectively, with shading indicating $\pm 1\sigma$ from the MLR regression coefficients. Black solid, dashed, and dotted curves in panel C show the three SST observation products for comparison.

Supporting Information for

Antarctic Ozone Depletion Shapes Surface Cooling Pattern and Climate Sensitivity

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This PDF file includes:

Figures S1 to S8
Tables S1

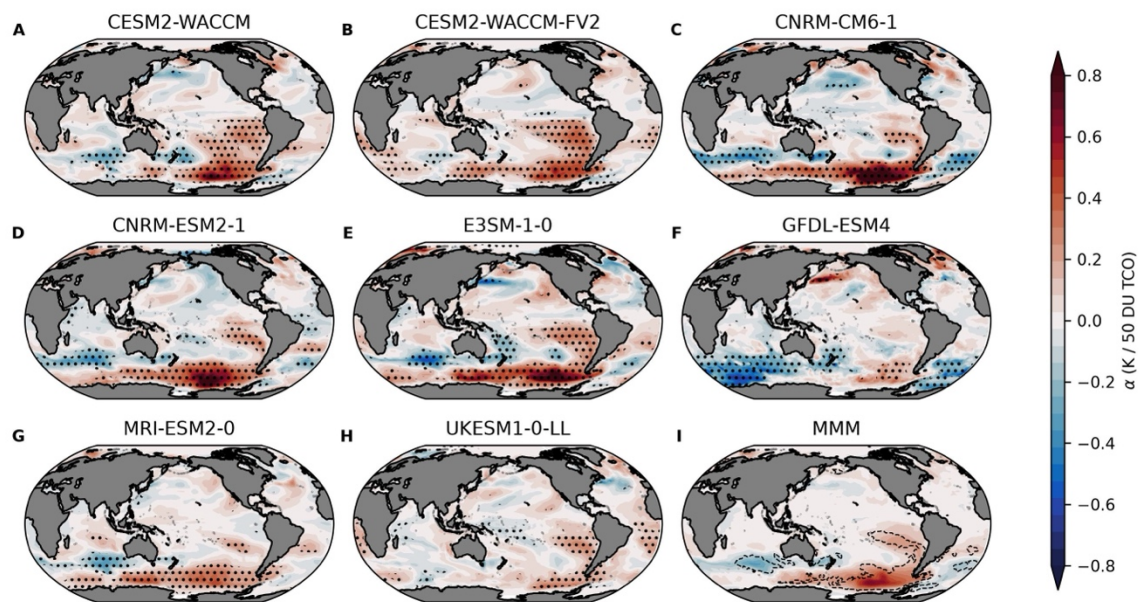


Fig. S1. Regression coefficients on Antarctic ozone in the MLR. Panels A-H show partial regression coefficients on Antarctic ozone for individual CMIP6 GCMs derived from their PiControl simulations. Dots indicate regions where the $p < 0.05$ in the MLR, denoting higher confidence in the SST response to Antarctic ozone. Panel I shows the multi-model-mean of the regression coefficients, with dashed contours highlighting regions where all eight GCMs agree on the sign of the SST response to Antarctic ozone.

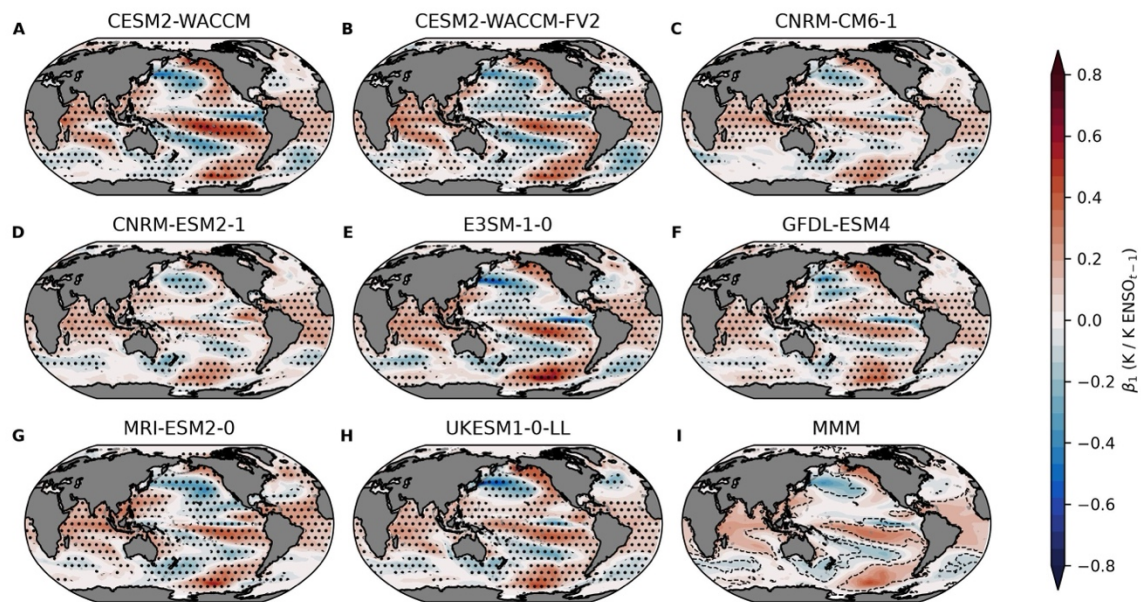


Fig. S2. Regression coefficients on ENSO (lag 1-year) in the MLR. Same as Fig. S1, but for ENSO with a 1-year lag.

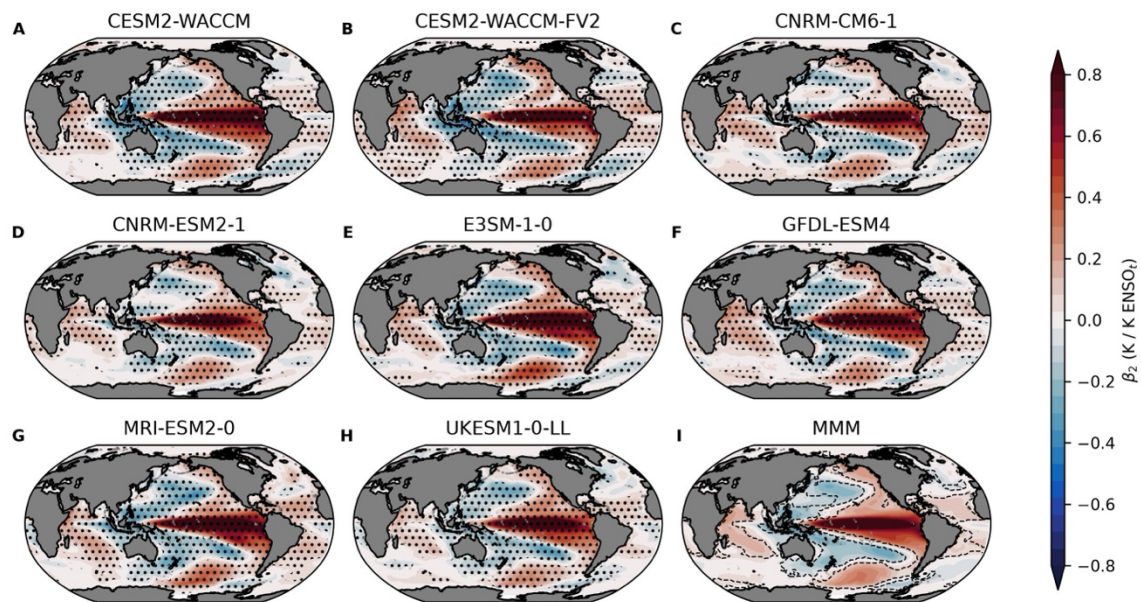


Fig. S3. Regression coefficients on ENSO (lag 0-year) in the MLR. Same as Fig. S1, but for ENSO with a 0-year lag.

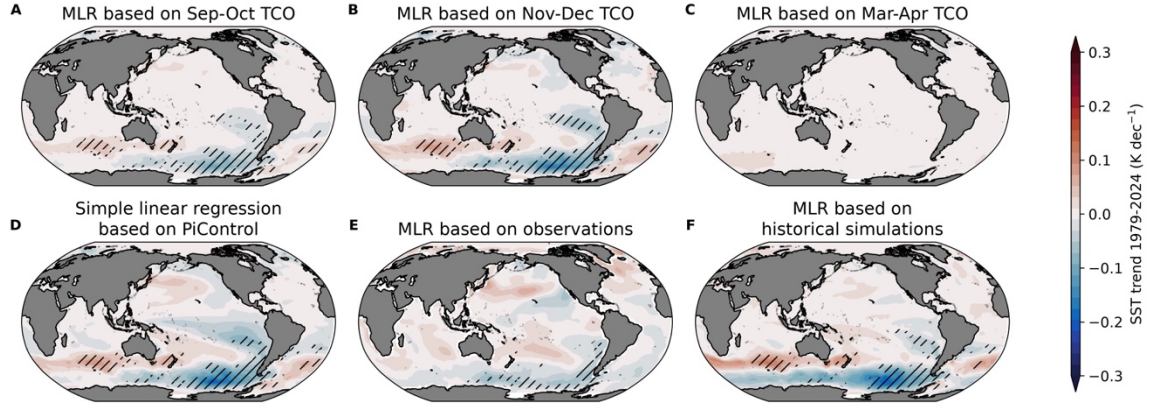


Fig. S4. Sea surface temperature trends from linear regression sensitivity tests. Panels A-C show SST trends during 1979-2024 derived from MLR using Antarctic ozone (60°S - 90°S) averaged over September-October, November-December, and March-April, respectively. Panel D shows SST trends derived from a simple linear regression using Antarctic ozone as the sole predictor (without including ENSO indices). Panels A-D are based on the multi-model mean of the regression coefficients from the eight CMIP6 PiControl simulations. Panel E shows the SST trend based on MLR using September-December Antarctic ozone constructed from observations during 1979-2022, with SST data from ERSST after subtracting the forced component estimated from ForceSMIP. Panel F shows the mean SST trend based on MLR using September-December Antarctic ozone constructed from internal variability in historical simulations during 1975-2014 for CNRM-CM6-1, CNRM-ESM2-1, MRI-ESM2-0, and UKESM1-0-LL. Hatching in regions where all eight PiControl regression coefficients agree on the sign of the response (dashed region in Fig. S1I) indicates where SST trends from these sensitivity tests lie within $\pm 1\sigma$ of the multi-model mean SST trend predicted by the PiControl MLR (Fig. 1C).

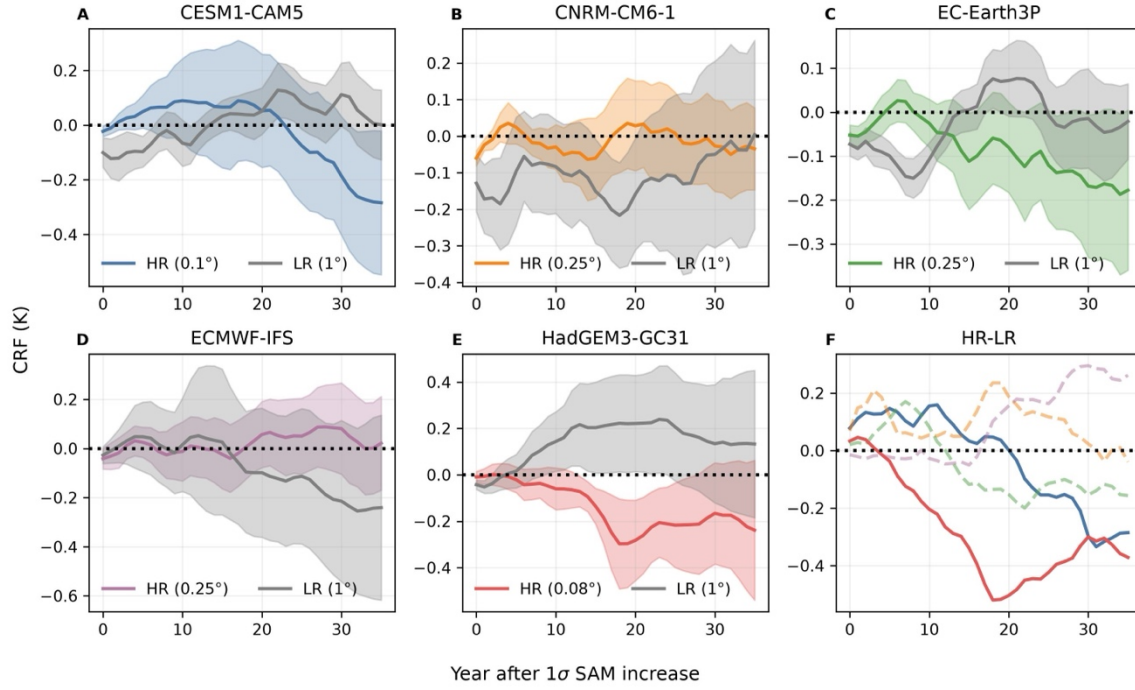


Fig. S5. Climate response function between high- and standard-resolution ocean models. CRFs for Southern Ocean SST in response to SAM increases are calculated from HighResMIP models with paired high- and standard-resolution configurations (panels A-E). Model horizontal resolutions are shown in the legend. Shadings indicate $\pm 1\sigma$ across a range of fitting parameter spaces discussed in Methods. The difference between the high- and standard-resolution CRFs is shown in panel F. The two GCMs with ocean resolutions finer than 0.1° (minimum resolution required to fully resolve mesoscale ocean eddies) are shown with solid curves, whereas the three GCMs with 0.25° ocean resolution are shown with lighter dashed curves in F.

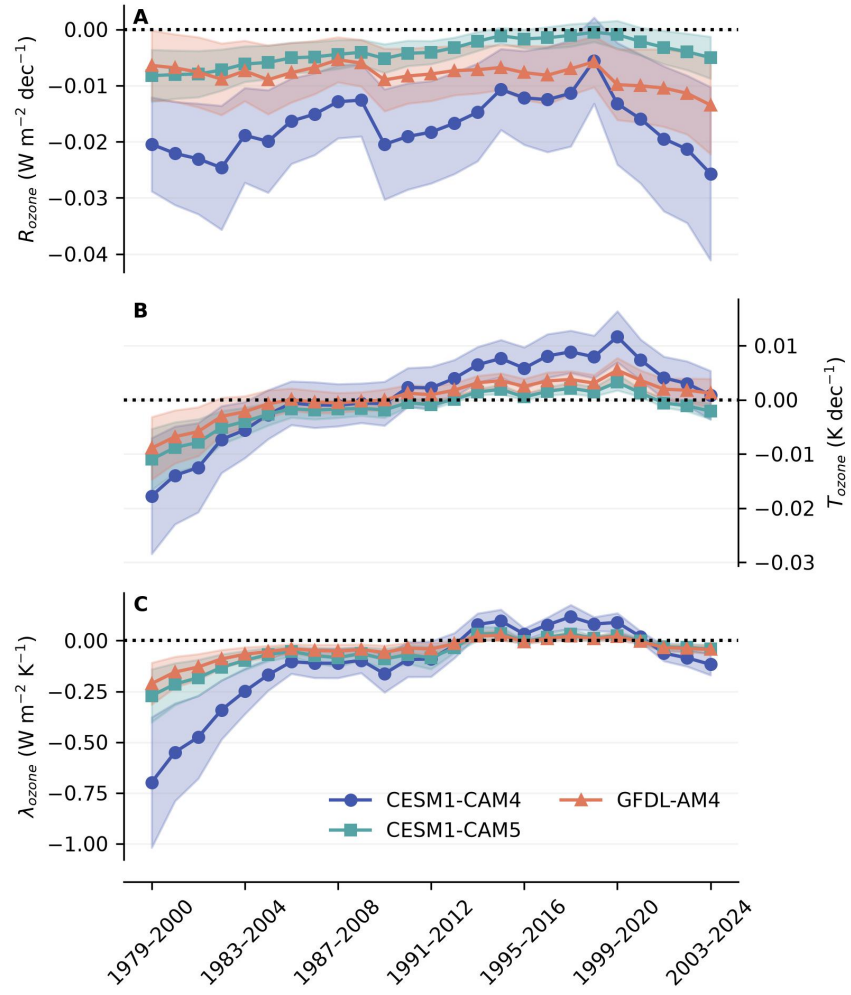


Fig. S6. Climate sensitivity due to ozone-induced sea surface temperature pattern changes.
Same as Fig. 3, but in absolute values.

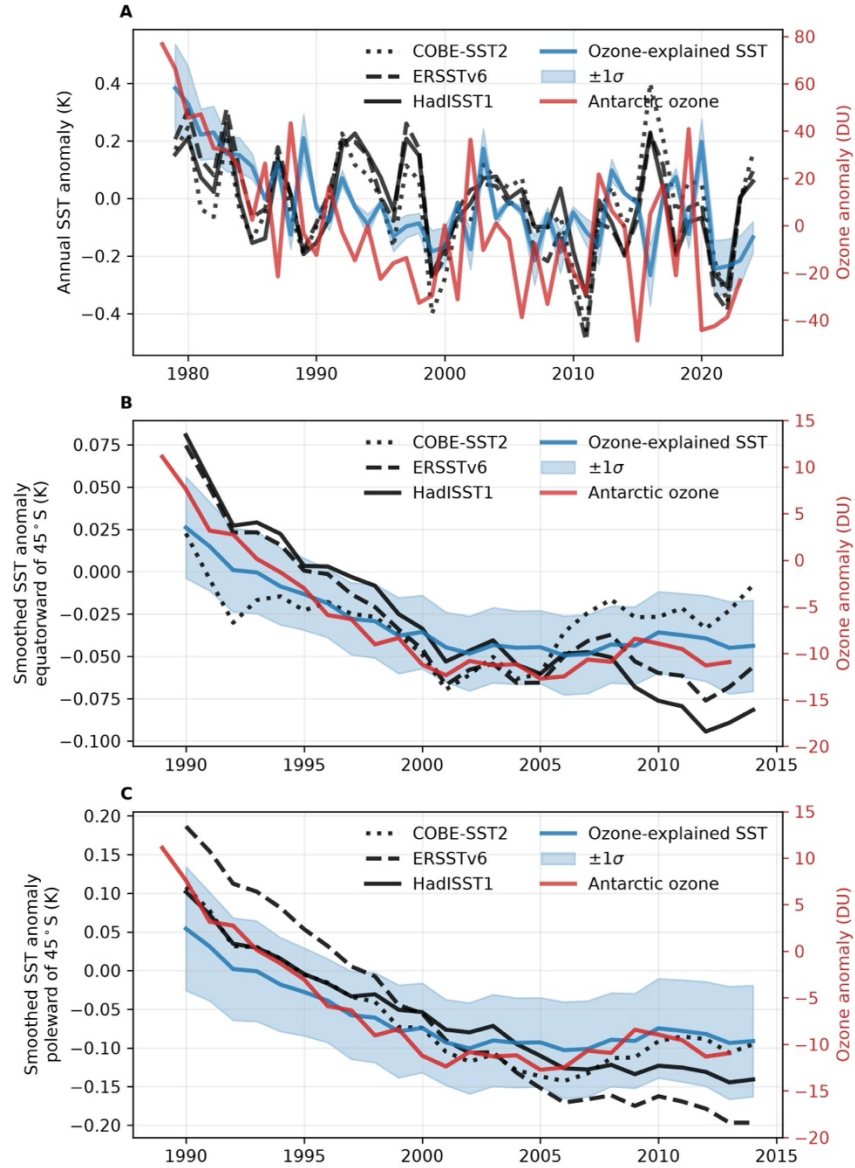


Fig. S7. Sea surface temperature anomalies on interannual timescales and regional averages. Panel A is the same as Fig. 4C but without the 22-year running mean. Panels B and C are the same as Fig. 4C but show the 22-year running-mean SST anomalies averaged separately over the eastern tropical Pacific (equatorward of 45°S) and the Southern Ocean (poleward of 45°S), respectively. The red curves (right y-axis) in all three panels show September-December total column ozone anomalies from the SBUV observations, averaged over 60°S-90°S.

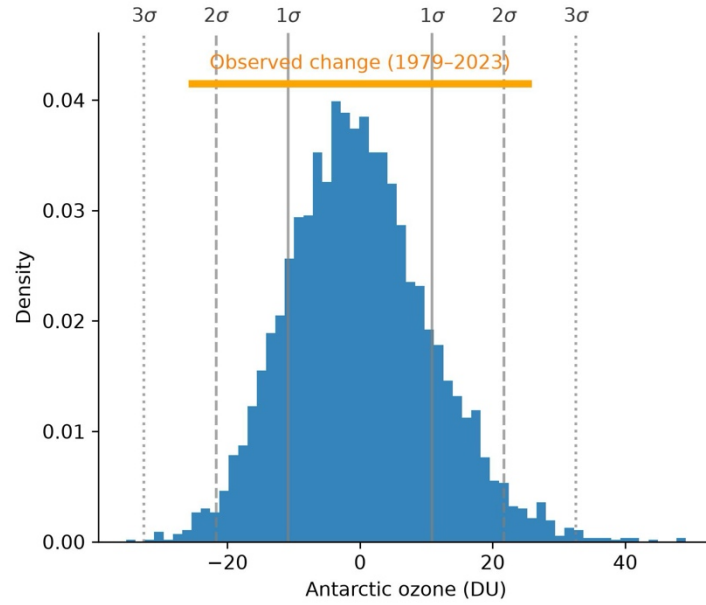


Fig. S8. Comparison of Antarctic ozone variability between PiControl simulations and observations. The histogram shows the distribution of September-December polar-cap (60°S-90°S) total column ozone anomalies in PiControl simulations from the eight CMIP6 GCMs considered in this study. The vertical gray solid, dashed, and dotted lines indicate the $\pm 1\sigma$, $\pm 2\sigma$, and $\pm 3\sigma$ ranges of PiControl variability, respectively. The horizontal yellow bar indicates the total change in ozone from SBUV satellite observations during 1979-2023. The observed Antarctic ozone change during the past four decades generally falls within the $\pm 2\sigma$ to $\pm 3\sigma$ range of ozone variability in the PiControl simulations.

Table S1. List of CMIP6 GCMs used in this study.

Model	Years of PiControl used to construct MLR	Historical and SSP availability	Stratospheric ozone
CESM2-WACCM	499	Historical+SSP2-4.5	Fully interactive
CESM2-WACCM-FV2		Historical	Fully interactive
CNRM-CM6-1		Historical+SSP2-4.5	Linearized scheme
CNRM-ESM2-1		Historical+SSP2-4.5	Fully interactive
E3SM-1-0		Historical	Linearized scheme
GFDL-ESM4		Historical+SSP2-4.5	Fully interactive
MRI-ESM2-0		Historical+SSP2-4.5	Fully interactive
UKESM1-0-LL		Historical+SSP2-4.5	Fully interactive